Book Market Analytics

Price and rating patterns on Books to Scrape

Fulufhelo Mulaudzi | 402601594 | Data Science 500 | 22 October 2026

1. Executive Summary

This project scraped **1,000 books across 50 genres** from Books to Scrape, cleaned them into an analytical dataset, explored pricing patterns and built a linear regression to predict price. Prices range from £10.00 to £59.99 (mean £35.07, median £35.98).

- **Rating and price:** the analysis finds no meaningful relationship between star rating and price (correlation = +0.03). A one-star increase in the average rating is associated with only a negligible shift in price.
- **Genre and price:** among genres with at least 10 books, the most expensive on average is **Travel** (£39.79) and the cheapest is **Food and Drink** (£31.41), a gap of £8.38 per title. A permutation test (p = 0.976) shows that a gap this large appears by chance when prices are shuffled across genres, so it should not be read as a real genre premium - see Section 3.
- **Prediction:** the final linear regression achieves a test R^2 of **-0.071** (no better than guessing the average), RMSE £15.00 and MAE £12.96 (5-fold CV R^2 -0.068), versus a mean-guess baseline RMSE of £14.53.
- **Implication:** the visible catalogue attributes explain almost none of a book's price, so they should not be used alone to set, judge or compare prices.

2. Methodology

- **Data acquisition:** BeautifulSoup scraper in two stages. First the 50 listing pages (title, price, star rating), then each book's own page (stock quantity, genre, description, UPC). A User-Agent header, polite delays between requests and try/except error handling were used.
- **Cleaning:** price text converted to numbers; rating words mapped to 1-5; stock quantity parsed from text; duplicate UPCs removed; missing descriptions kept (flagged with a has_description column) rather than dropping rows that still had valid prices. Final dataset: 1,000 rows and 15 columns.
- **Exploration:** descriptive statistics, histograms, box plots, scatter plots, genre comparisons (with a permutation test for the genre gap) and a correlation matrix.
- **Modelling:** 80/20 train-test split (random_state=42); genre one-hot encoded with a reference category dropped so genre coefficients are interpretable as £ differences versus that reference; baseline vs numeric-only vs numeric + genre models; evaluated with R^2, RMSE, MAE and 5-fold cross-validated R^2.

3. Key Insights

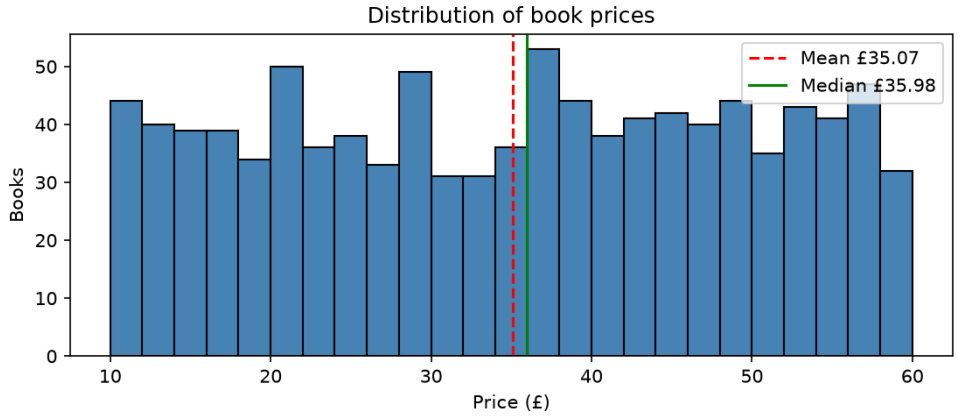


Figure 1: price distribution

Price skewness is -0.04: the distribution is approximately symmetric. Prices are spread fairly evenly across the whole range, so the catalogue serves budget, mid-range and premium buyers alike.

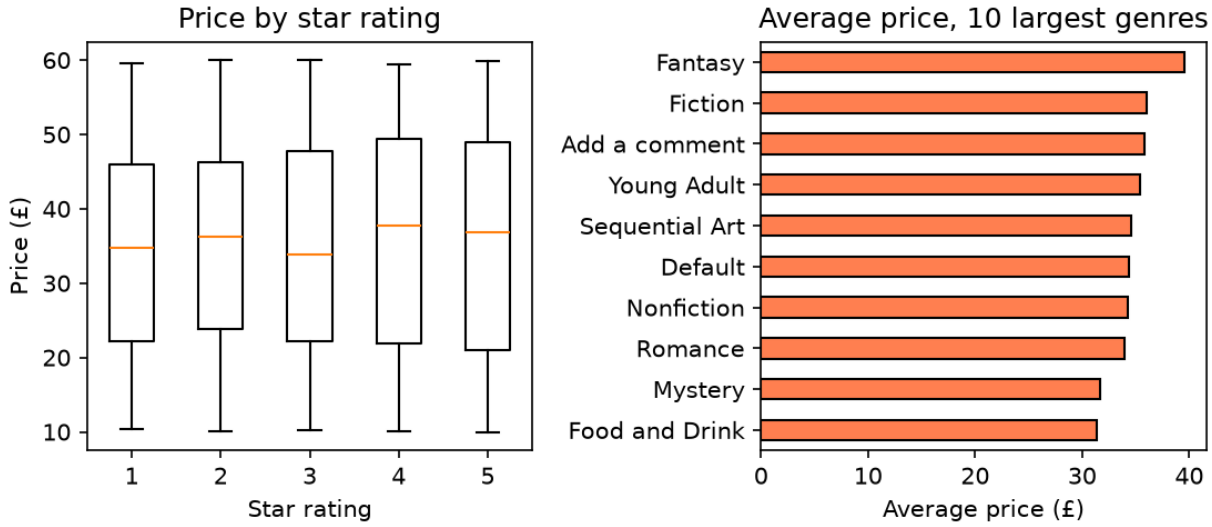


Figure 2: price by rating (left) and average price by genre (right)

Rating is not a price signal. The box plots in Figure 2 show almost complete overlap between the price ranges of 1-star and 5-star books, and the correlation between rating and price is only +0.03. Stock quantity has a correlation of -0.01 with price, which is effectively no relationship. Retailers and publishers should not use either rating or stock level as evidence that a title can command a higher list price.

Genre gap: Travel has the highest average price (£39.79) and Food and Drink the lowest (£31.41), a difference of £8.38 per title. A permutation test shuffled the prices across genres 2000 times; a gap this large appeared in 97.6% of shuffles ($p = 0.976$), so it is the kind of difference chance produces when many genres are compared. Both genres contain only about 11 books, so this gap should not be read as a genuine genre premium. The catch-all labels (Default, Add a comment) were left out of this comparison.

4. Modelling Outcomes

Model	Test R2	RMSE (£)	MAE (£)	CV R2
Baseline (predict mean)	-0.005	14.53	12.66	-0.003
Linear Regression: numeric only	-0.018	14.62	12.76	-0.013
Linear Regression: numeric + genre	-0.071	15.00	12.96	-0.068

The final model (numeric features + genre) does not meaningfully improve on the baseline. The largest coefficients belong to **genre_Business** (-7.18), **genre_Mystery** (-6.86), **genre_Childrens** (-5.70). A train R^2 of 0.028 against a test R^2 of -0.071 indicates limited over-fitting.

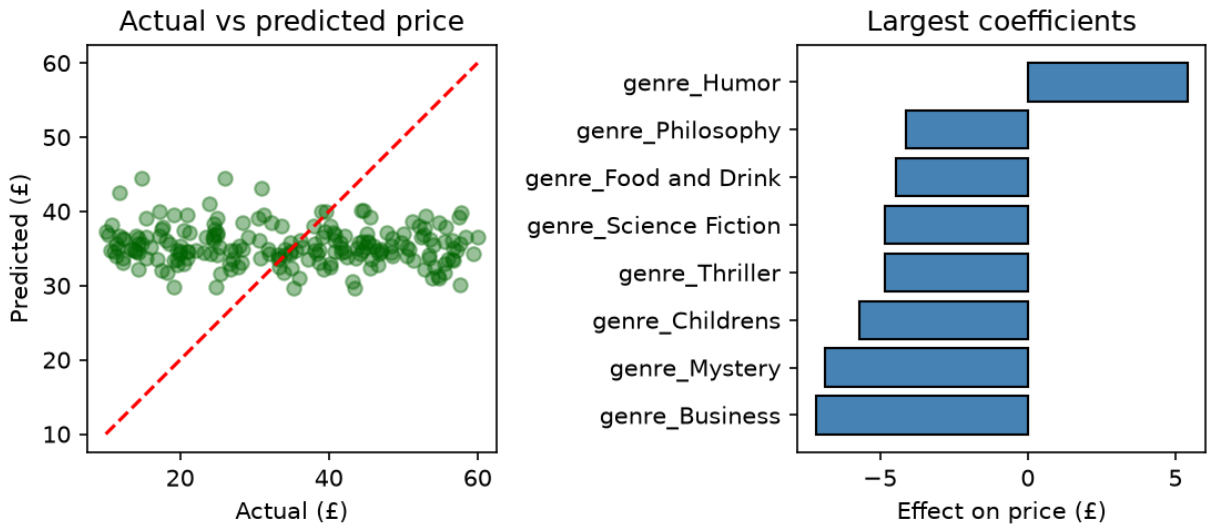


Figure 3: actual vs predicted price (left) and largest coefficients (right)

Limitations

- Linear model only: no curved relationships or interactions.
- Key real-world price drivers (author, publisher, format, page count, publication date) are not on the site.
- Books to Scrape is a sandbox: its prices are not set by a real market, so results show the method, not true bookselling economics.
- Star rating is ordinal but treated as numeric; many genres have only about 20 books, so genre effects are noisy.

5. Recommendations

- **Retailers:** do not use star rating or stock level as pricing signals; they carry almost no information here. Genre differences were no larger than chance, so do not reprice by genre on this evidence. Base shelf prices on supplier cost, target margin and observed competitor prices, and test any new price on a small scale before a full roll-out.
- **Publishers:** a high star rating does not go with a higher price on this site, so a well-rated title should not be assumed to justify a premium list price. To build a pricing model that works, collect richer attributes such as author history, page count, format and publication date, none of which appear on Books to Scrape.
- **Data-driven buyers:** a higher price does not go with a higher star rating on this catalogue, so compare ratings and descriptions directly instead of using price as a shortcut for quality.

- **Next step:** the final model's test R^2 is -0.071, so the available attributes explain almost none of the price variation. Adding author, publisher and page count is a sensible next step, but this notebook cannot show how much they would improve the predictions.

6. Conclusion

The results show that on this sandbox site, star rating and stock level do not meaningfully predict book price (R^2 close to zero). For a retailer, this means pricing decisions should be based on supplier cost, margin targets and competitor prices rather than customer ratings. Publishers should not assume that a highly-rated title can command a higher list price without further evidence from a real market.